

VAIBHAVI MUTYA

Data Engineer | Data Analytics | AI-Powered Solutions
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PROFESSIONAL SUMMARY

Senior Data Engineer with 5+ years of experience building scalable data pipelines, lakehouse architectures, and AI-powered analytics across financial services and healthcare. Proficient in Python, PySpark, SQL, AWS, Azure, GCP, and Databricks, with hands-on expertise in RAG/LLM integration, anomaly detection, and MLOps. Proven track record of reducing manual effort, improving data quality, and delivering actionable insights for risk, compliance, and operations teams.

TECHNICAL SKILLS

Languages: Python, SQL (PostgreSQL, Redshift, MySQL), R

Data Engineering & Modeling: ETL/ELT Pipelines, Data Modeling, Azure Synapse Analytics, Snowflake, Apache Spark (PySpark), Apache Flink, Apache Iceberg, Kafka, dbt, Airflow, Azure Data Factory, AWS Glue, Delta Lake, DynamoDB, Data Lakehouse, Data Mesh, DataOps, Time Series Forecasting, Clustering, Regression Analysis

AI & Machine Learning: LLM Integration, RAG (Retrieval Augmented Generation), Semantic Search, Vector Databases, Generative AI, Amazon Bedrock, Data Anomaly Detection, Predictive Analytics

Business Intelligence & Visualization: Amazon QuickSight, Power BI, Tableau, Looker Studio, Microsoft Excel

Statistical & Analytical Techniques: T-test, Chi-squared Analysis, KPI Tracking, Causal Impact Analysis, A/B Testing

Cloud & DevOps: Azure (Data Factory, Synapse Analytics, Blob Storage, OpenAI Service, DevOps), AWS (S3, Redshift, Glue, Bedrock, OpenSearch), Google Cloud (BigQuery, Cloud Storage), Databricks, Docker, Kubernetes, Terraform, CI/CD Pipelines

Version Control & Methodology: Git, GitHub, Agile/Scrum

Automation & Scripting: Python Automation, VBA, Audit Frameworks, Reporting Automation

EXPERIENCE

Senior Data Engineer

Jun 2025 – Present

Northern Trust, USA

- Designed **AWS-based predictive analytics pipelines** using **Python, SQL, AWS Glue, Amazon S3, and Amazon Redshift** to process **20M+** transaction records per month, improving intraday reporting for risk, compliance, and operations teams.
- Built production-ready **machine learning workflows** using **Apache Spark (PySpark), Python, and AWS services** to support anomaly detection, transaction monitoring, and operational risk analysis, reducing manual review effort by approximately **8–10 hours per week** and improving investigation consistency.
- Developed **NLP, semantic search, and RAG-enabled analytics workflows** using **Amazon Bedrock and vector databases** to accelerate policy, compliance, and transaction insight retrieval, reducing research turnaround time by **25%**.
- Engineered scalable **data lakehouse workflows** on **Amazon S3** using **Delta Lake and Apache Spark (PySpark)** to build model-ready datasets for analytics and BI use cases, reducing data preparation effort by **30%**.
- Implemented **SQL and Python-based data quality, validation, and audit checks** integrated with **Amazon OpenSearch** to identify missing, duplicate, and inconsistent records, reducing recurring validation exceptions by **15%** and strengthening reporting reliability.
- Partnered with risk, compliance, and business stakeholders to translate **model outputs, RAG insights, and anomaly patterns** into actionable recommendations, improving issue resolution cycle time by **20%**.

Associate Data Engineer

Apr 2023 – May 2025

Northern Trust, USA

- Developed **Azure-based predictive data pipelines** using **Python, SQL, Azure Data Factory, Azure Synapse Analytics, and Azure Databricks** to forecast back-office staffing from transaction volume, improving capacity planning accuracy by **10%**.
- Built **machine learning-ready feature datasets** from multi-year transaction history using **Apache Spark (PySpark) and Azure Data Lake Storage**, enabling causal impact analysis for policy, fee, and operational change evaluations across **3+ years** of historical data and supporting leadership reporting.
- Supported **MLOps-style deployment workflows** through **Azure DevOps, GitHub, and CI/CD pipelines**, reducing deployment rework by **18%** and improving release traceability across dev and production environments.
- Implemented **Azure-based data governance controls** across curated data layers using audit logic, validation rules, and metadata-driven workflows, improving compliance reporting readiness and reducing manual follow-up effort.
- Collaborated with business and operations teams to convert **statistical findings, forecasting outputs, and KPI trends** into clear recommendations, reducing manual reconciliation checks by **25%**.

NTT DATA, India

- Developed healthcare analytics dashboards using **BigQuery, Looker Studio, Power BI, and Tableau** to monitor patient service KPIs, operational SLA performance, and claims-related reporting trends, enabling daily visibility across **5+ operational workstreams** for healthcare support teams.
- Automated healthcare data collection and reporting workflows using **Python, SQL, Google Cloud Storage, BigQuery, and Azure Blob Storage**, reducing recurring manual reporting effort by approximately **6 hours per week** while improving report repeatability and data consistency.
- Built **predictive analytics and statistical models** using **Python, SQL, clustering, regression analysis, t-tests, and chi-squared analysis** to identify healthcare operational trends and support supplier planning, inventory control, process improvement, and service performance analysis.
- Designed **SQL-based anomaly detection checks** for healthcare logistics, patient service, and operational datasets using **BigQuery and Azure SQL**, improving early identification of missing, duplicate, or inconsistent records before monthly audit reviews and compliance reporting cycles.

EDUCATION

University of California, Riverside

Master of Science in Computer Science

Riverside, CA

Dec 2022

GITAM University

Bachelor of Technology in Computer Science Engineering

India

Jun 2020

CERTIFICATIONS

Databricks Certified Data Engineer Associate | Databricks

Generative AI with Large Language Models | DeepLearning.AI & AWS

AWS Certified Data Engineer – Associate | Amazon Web Services

Google Professional Data Engineer | Google Cloud

PROJECTS

AI-Powered Financial Document Analysis Using RAG & LLMs

- Developed a **Retrieval Augmented Generation (RAG)** system using **LangChain, OpenAI GPT-4, and FAISS vector database** to extract and analyze key financial metrics from **500+ 10K SEC filings** and earnings reports for Fortune 500 companies, achieving **85% accuracy** vs. manual analysis.
- Engineered a **semantic chunking and embedding pipeline** using **sentence-transformers** to index financial documents into a vector store, enabling sub-second similarity search and contextual retrieval across large document corpora.
- Implemented **prompt engineering and retrieval tuning** to reduce LLM hallucination rate by **40%**, validated against human expert annotations; deployed interactive **Streamlit** dashboard for natural language financial queries.
- **Technologies:** Python, LangChain, OpenAI API, FAISS, Hugging Face Transformers, Streamlit, Pandas, BeautifulSoup

Earthquake Analytics End-to-End Data Pipeline & Dashboard

- Built a production-style **end-to-end data pipeline** using **Python and USGS API** to ingest, clean, and transform real-time earthquake GeoJSON data; applied feature engineering to extract **temporal attributes, magnitude tiers, depth categories, and regional metadata** for downstream analytics.
- Developed a **Streamlit interactive dashboard** with country-level filtering, KPI cards, and geospatial map visualization; containerized the full application with **Docker** for portable, reproducible deployment.
- Created **multi-dimensional visualizations** using **Plotly and Matplotlib** including magnitude-depth scatter plots, monthly trend charts, and regional bar charts to support exploratory seismic data analysis.
- **Technologies:** Python, Streamlit, Pandas, NumPy, Plotly, Matplotlib, Requests, Docker, USGS GeoJSON API